

Group 5.B: Simon says game

By :

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Game Principle

PHONE SHOWS INSTRUCTION

"Simon says RUN_IN_PLACE"

"Simon says JUMP"

"SAY_HI" (trap!)



WATCH DETECTS PLAYER GESTURE

Accelerometer + Gyroscope

50Hz sampling

CNN+LSTM ML Model

Classification

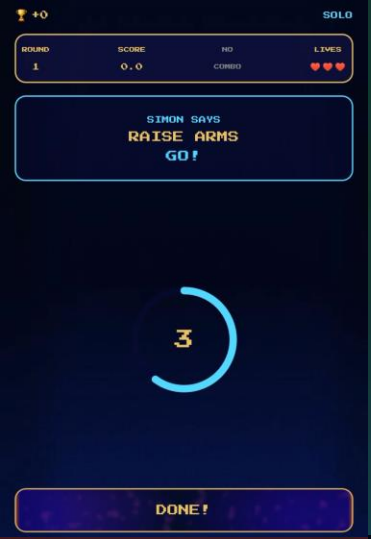
Scoring System

Points = 1 × Multiplier × (1 + SpeedBonus)

SCENARIO	SIMON SAYS?	PLAYER ACTION	RESULT
Correct Move	YES	Did expected	**+ additional points**
Wrong Move	YES	Did different	**combo reset**
Avoided Trap	NO	Did different or nothing (IDLE)	**+ additional points**
Fell for Trap	NO	Did expected	**-1 point**

Speed Bonus: Response time vs allowed time (faster = more points)

Combo: Consecutive successes multiply points (2×, 3×, 4×...)


+0
SOLO
ROUND
SCORE
NO
LIVES
1
0.0
COMBO
SIMON SAYS
RAISE ARMS
GO!
3
DONE!


+1
SOLO
ROUND
COMBO
LIVES
9
11.2
×2
GREAT!
5.0s | +0% SPEED
+2.0 pts
TROPHY +1
CONTINUE
QUIT


+0
SOLO
ROUND
SCORE
COMBO
LIVES
8
6.7
×2
GREAT!
3.5s | +24% SPEED
+2.5 pts
+2.5
QUIT

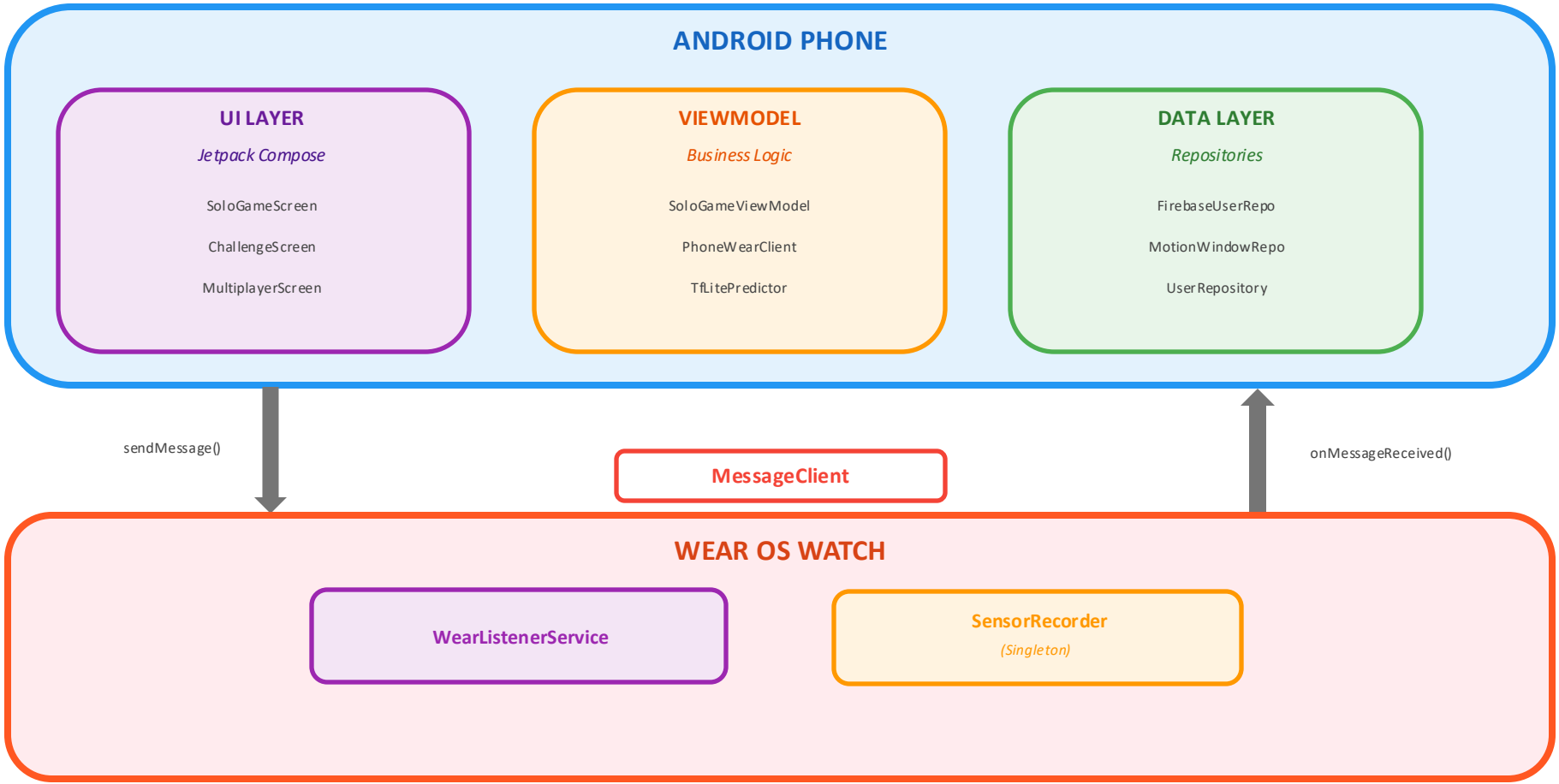

+1
SOLO
ROUND
SCORE
NO
LIVES
13
15.2
COMBO
IDLE
You didn't move!
DEBUG PROBAS
Expected: JUMP
IDLE
100%
✗ WRONG
LIVES REMAINING:
CONTINUE
QUIT


+1
SOLO
ROUND
SCORE
NO
LIVES
14
15.2
COMBO
MISSED!
GAME OVER
DEBUG PROBAS
Expected: JUMP
✗ WRONG
CONTINUE
QUIT

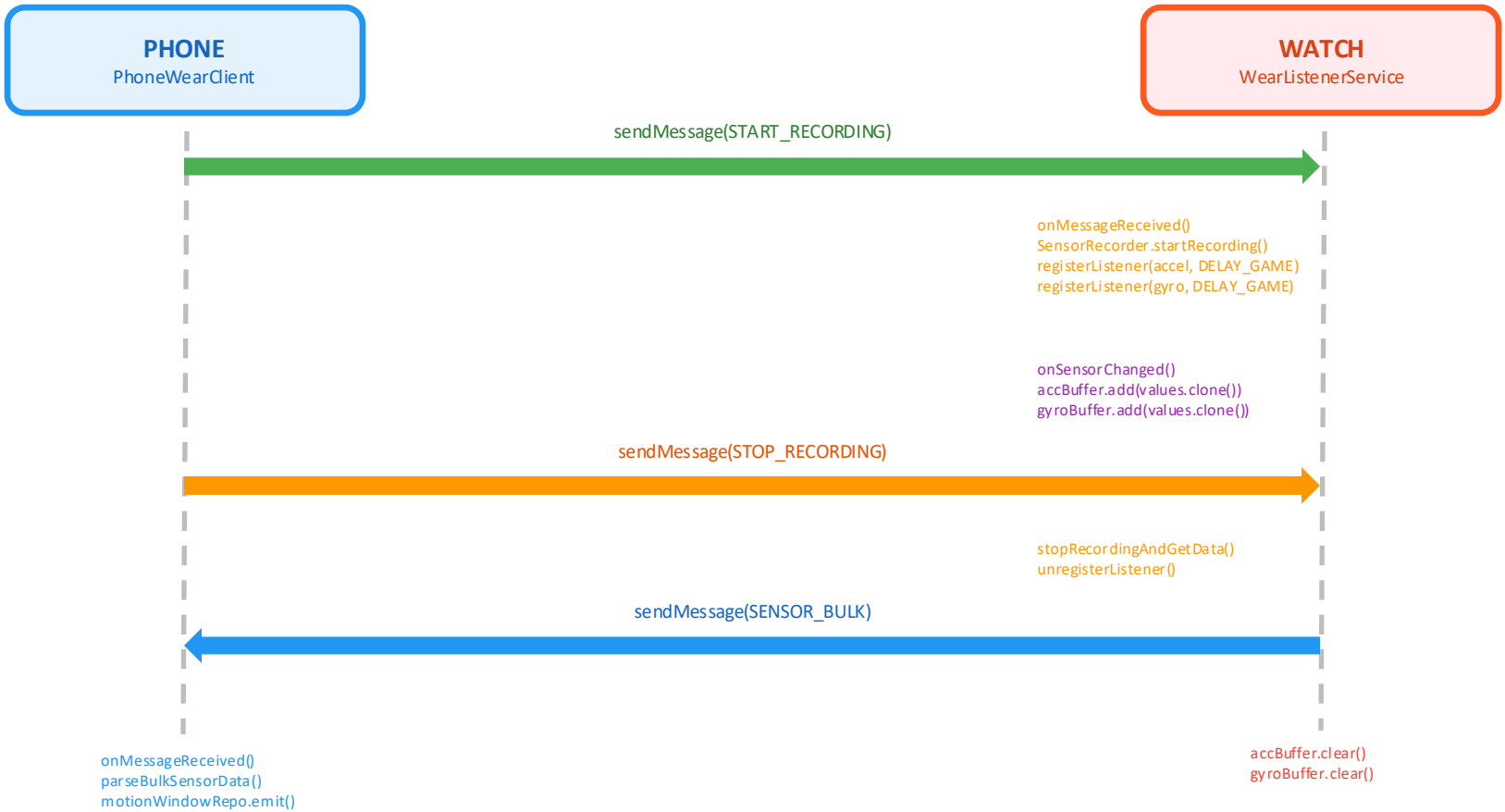

+0
SOLO
ROUND
SCORE
NO
LIVES
6
6.7
COMBO
TRAP!
I didn't say
SIMON SAYS!
-1 POINT
-1
QUIT


+0
SOLO
ROUND
SCORE
NO
LIVES
1
1.0
COMBO
GREAT!
5.0s | +0% SPEED
+1.0 pts
DEBUG PROBAS
Expected: RAISE_ARMS
RAISE_
40%
SAY_HI
15%
RUN_IN
15%
SPIN_A
15%
JUMP
13%
CLAP
0%
✓ CORRECT
CONTINUE
QUIT

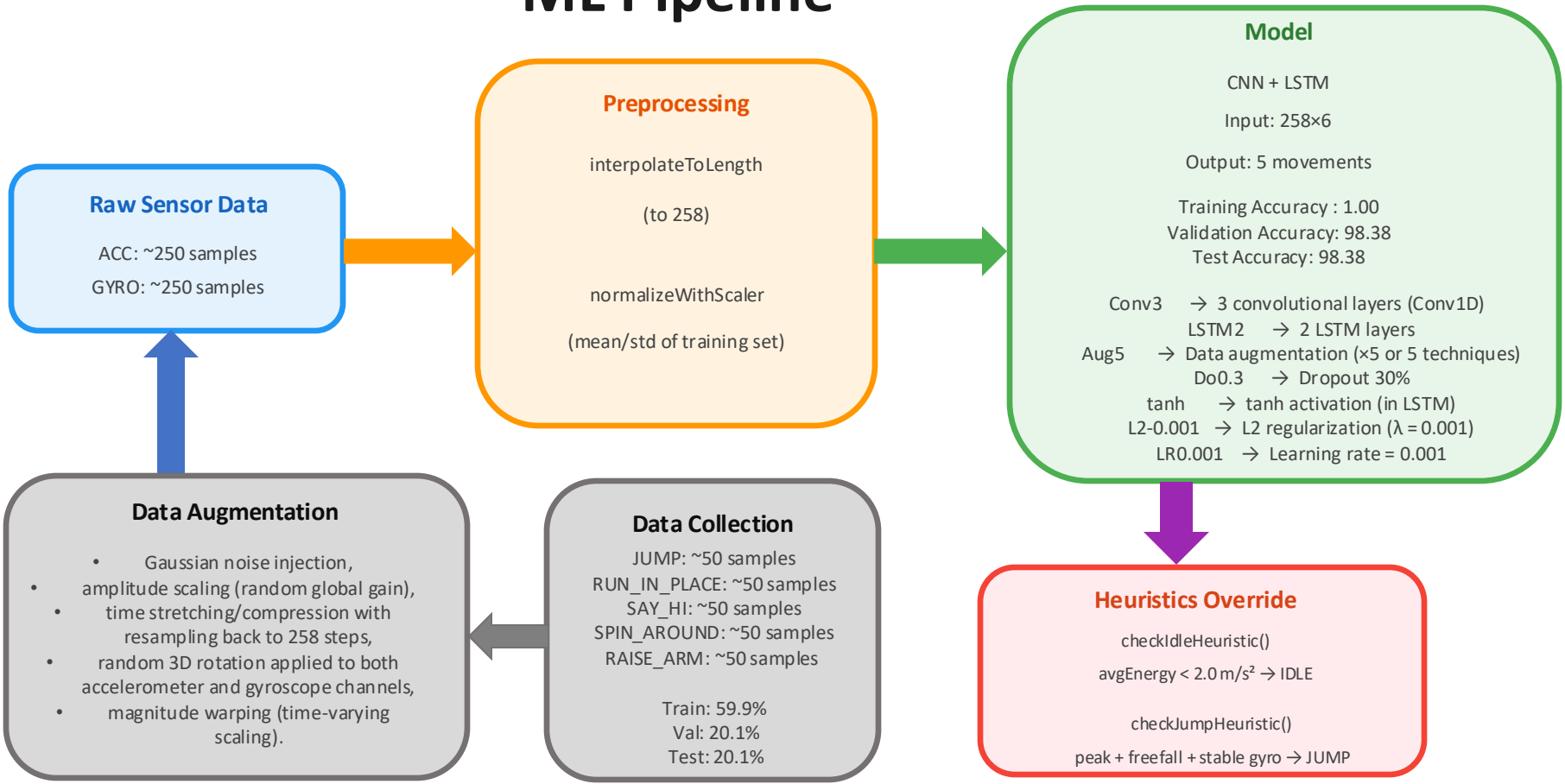
System Architecture



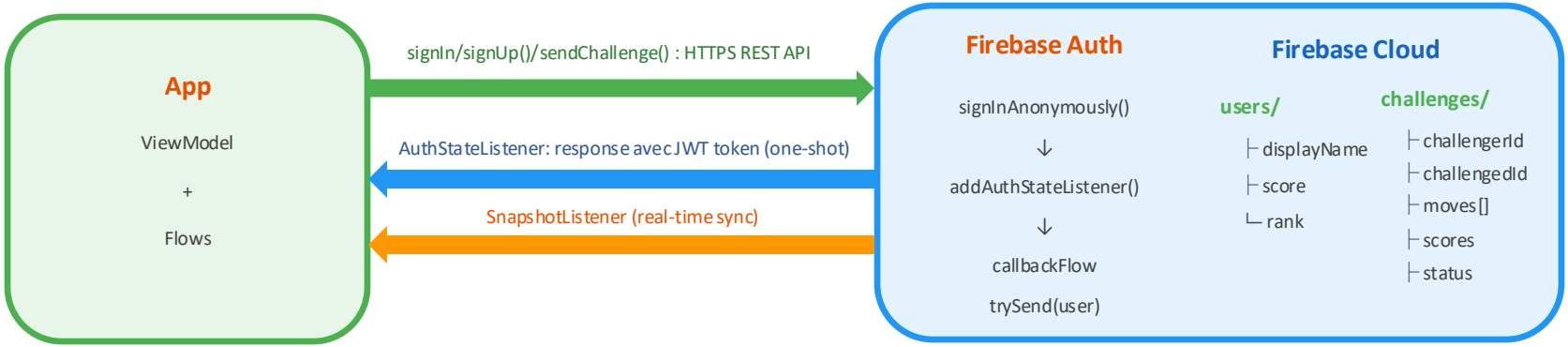
Phone-Watch Communication Protocol

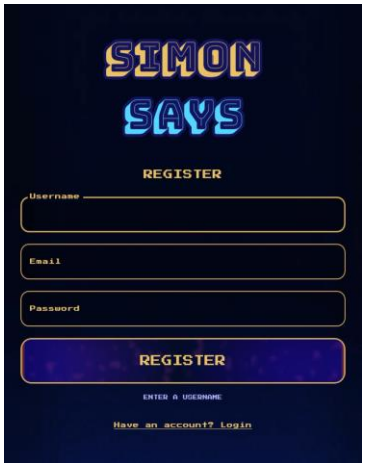


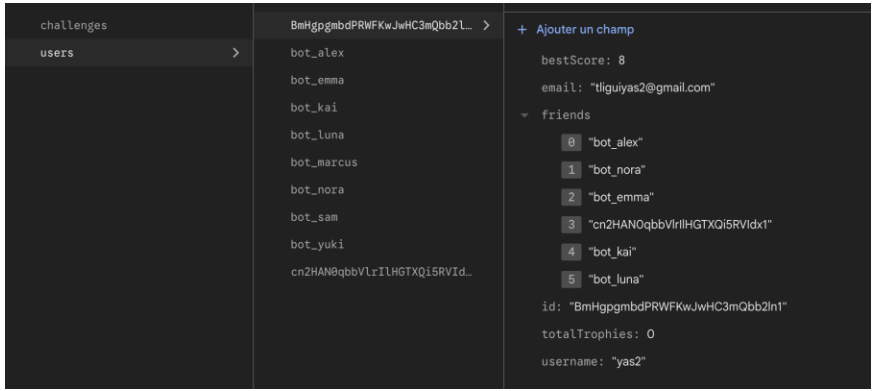
ML Pipeline


Raw Sensor Data
ACC: ~250 samples
GYRO: ~250 samples
Preprocessing
interpolateToLength
(to 258)
normalizeWithScaler
(mean/std of training set)
Model
CNN + LSTM
Input: 258×6
Output: 5 movements
Training Accuracy : 1.00
Validation Accuracy: 98.38
Test Accuracy: 98.38
Conv3 → 3 convolutional layers (Conv1D)
LSTM2 → 2 LSTM layers
Aug5 → Data augmentation (×5 or 5 techniques)
Do0.3 → Dropout 30%
tanh → tanh activation (in LSTM)
L2-0.001 → L2 regularization (λ = 0.001)
LR0.001 → Learning rate = 0.001
Heuristics Override
checkIdleHeuristic()
avgEnergy < 2.0 m/s² → IDLE
checkJumpHeuristic()
peak + freefall + stable gyro → JUMP
Data Collection
JUMP: ~50 samples
RUN_IN_PLACE: ~50 samples
SAY_HI: ~50 samples
SPIN_AROUND: ~50 samples
RAISE_ARM: ~50 samples
Train: 59.9%
Val: 20.1%
Test: 20.1%
Data Augmentation
• Gaussian noise injection,
• amplitude scaling (random global gain),
• time stretching/compression with resampling back to 258 steps,
• random 3D rotation applied to both accelerometer and gyroscope channels,
• magnitude warping (time-varying scaling).

Firebase Integration











Multiplayer Offline Mode

Challenge Mode





Conclusion

WHAT WE BUILT
- Simon Says game with real-time gesture recognition
- Phone + Smartwatch communication via Wearable API

KEY TECHNOLOGIES
- TensorFlow Lite CNN+LSTM model (5 gestures, ~98% accuracy)
- Firebase Auth + Firestore (real-time sync)
- Offline challenge system between players

CHALLENGES SOLVED
- Sensor data synchronization (50Hz, 258 samples)
- Jump detection heuristic (freefall + landing peak)

POSSIBLE IMPROVEMENTS
- Add more gestures
- Multiplayer real-time mode